The Epilog Fusion Edge

24

Operation, safety and troubleshooting for the 80 W CO_2 laser cutter. The teaching chapters, diagrams, self-check answers and manufacturer references.

LAB	DATE	VERSION
Fab Lab Saga, University of Iceland	2026-10-21	80 W CO_2 · Model 17000 · September 2021 · 50.8 mm lens

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Manufacturer references are listed in Maintenance → Sources.

Safety rules

Ten rules, how to stop and how to start a job. Read them before the induction.

1 **Trained and signed off.**
You run the laser only after the induction and this self-check. Students and staff operate it; cleaning, supervision, service and release are staff work.

2 **Never leave it running.**
Stay at the machine and watch the whole job. A camera, a timer or someone passing by does not count. Vector cutting is the most likely to start a fire, acrylic especially. [2,5]

3 **Identify the material.**
If you can't name the material, it doesn't go in. PVC and vinyl never: they release a corrosive gas that destroys the machine and harms you. Kydex contains PVC. If in doubt, read its safety data sheet or ask staff. [2,6,7]

4 **Measure, then enter the thickness.**
Use calipers. Enter the thickness for every process. Left blank, autofocus focuses on the top of the grid instead of your material. [2]

5 **Extraction on, always.**
Never run without working extraction. At Saga it runs all the time; if you smell smoke at the lid, or smoke hangs in the cabinet, pause and tell staff. [2]

6 **Lid shut while it runs.**
The lid stops the beam, not the head. Pause with Go/Stop and wait for the head to stop before you lift it. [2]

7 **Flame: pause, step back, call.**
Flicker is normal; flame that grows is not. Press Go/Stop, keep the lid shut and your hands out, and get staff. [2,5]

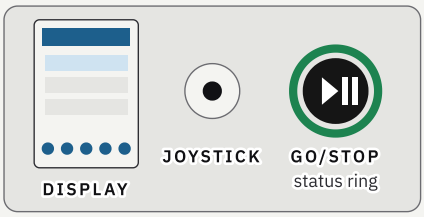
8 **Leave the bed clean.**
Offcuts that fall through the grid are fuel. Take out your parts and scraps; tell staff if the tray needs emptying. [2,5]

9 **Hands off the optics and panels.**
Never run with a cover off or the enclosure changed. Lens, mirrors and panels are staff work, with approved cleaner only. [2]

10 **Check the load limit.**
27.2 kg at most while the table moves (autofocus moves it), 40.8 kg resting. Heavy or awkward work is staff work. [13]

How to stop

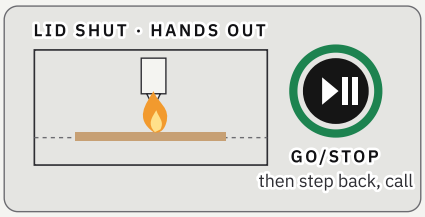
Go/Stop is the only stop control; there is no emergency-stop button.



ROUTINE · THE JOB CAN RESUME

Go/Stop

The round button by the display. The beam shuts off at once. The head pauses at the end of the raster line, or at the next node of a vector path, so it may keep moving for a moment. Then you may lift the lid and look. Lid down and Go/Stop again resumes where it left off. It lines up only if nothing moved. [2]



SOMETHING IS WRONG · DON'T RESUME

Flame, smoke, a smell

Press Go/Stop, keep the lid shut, don't reach in, step back and get staff. Lifting the lid stops the beam but not the head. [2]

How to start a job

Enter the thickness before you print; run Trace before Go/Stop.

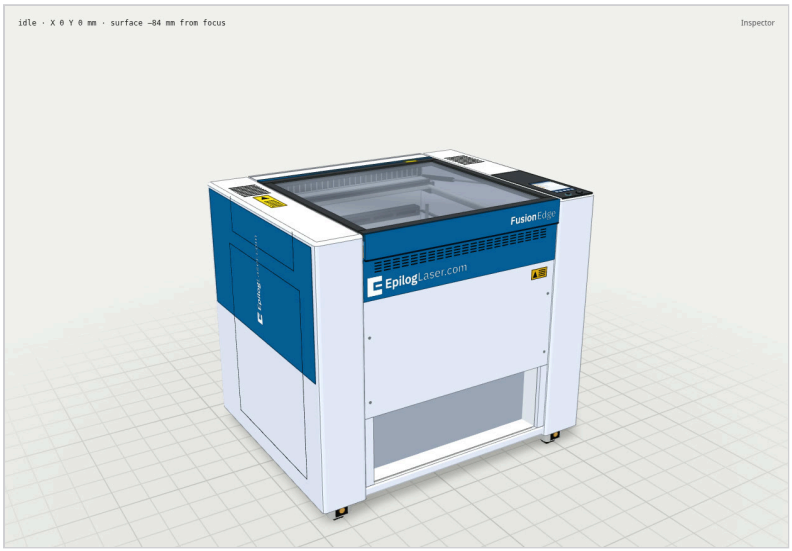
1. **Turn it on**
   Turn the machine on and wait until the boot cycle is complete. Check that air assist is on.
2. **Print to the laser**
   In the Epilog software: Auto Focus = Thickness, thickness entered for every process.
3. **Select your job**
   Job Menu on the panel; your file name highlighted in blue.
4. **Trace**
   Red dot on, Trace: the head outlines the job. Check that it lands on the sheet.
5. **Lid down, look**
   Nothing loose on the bed, you in place for the whole job.
6. **Go/Stop**
   The table rises to focus, then the job runs. Watch every second.

Epilog safety list (manual and Support)

- Never leave the laser running unattended. [2]
- Vector cutting is the most likely to start a fire; acrylic especially. [2]
- Keep the area clear of clutter, combustibles and solvents. [2]
- Always use air assist when vector cutting. [2]
- Never run without working extraction. [2]
- Some materials (paint, varnish, composition board, plastics) give off harmful compounds. [2]
- Never engrave or cut anything containing PVC or vinyl. [2]
- Never run with covers removed or the enclosure modified. [2]
- Access panels are not interlocked; never run with one removed. [2]
- As built, the system is a Class 2 laser product. [2]

The machine

Beam path, motion system, table and extraction: what each part does.



WORK AREA	610 × 610 mm
LASER	80 W CO_2, 10.6 µm, air-cooled
LENS	50.8 mm focal length
MAX THICKNESS	254 mm
TABLE LOAD	27.2 kg moving · 40.8 kg resting
MACHINE	1055 × 834 × 967 mm · 102 kg
EXTRACTION	595–680 m^3/h, one 102 mm port

WORK AREA	LASER	LENS	MAX THICKNESS	TABLE LOAD	MACHINE	EXTRACTION
610 × 610 mm	80 W CO_2, 10.6 µm, air-cooled	50.8 mm focal length	254 mm	27.2 kg moving · 40.8 kg resting	1055 × 834 × 967 mm · 102 kg	595–680 m^3/h, one 102 mm port

The model. Outer dimensions (1055 × 834 × 967 mm) and work area (610 × 610 mm) are to scale. [1] [17] The inside is simplified. The CO_2 beam is drawn red so you can see it; the real beam is invisible.

Mechanisms

`cabinet`	A closed sheet-metal cabinet on a built-in pedestal with four wheels. It fully encloses the laser and its beam path; two interlocks switch the beam off when the lid opens. The access panels are **not** interlocked, so they stay on. [1,2]
`source`	An air-cooled CO_2 tube in the rear bay. Saga's unit is 80 W. Its light is 10.6 µm: infrared, invisible, and it doesn't pass through the window. [4]
`beam path`	Four mirrors and one window optic carry the beam forward, across the X-beam and down into the head. The last mirror turns it down through the lens. Dirty optics lose power and get damaged. [2,11]
`head`	The lens focuses the beam to a small spot a fixed distance below the cone. Saga's machine has the standard 50.8 mm lens. Cone air assist blows through the head whenever a job runs, keeping the lens clean and flames down. [1,2]
`motion`	Brushless DC servo motors with encoders move the X-beam (Y) and the head along it (X), on Kevlar belts. The table moves in Z: it rises until your material's surface sits at focus. [2,4]
`red dot`	A visible red laser runs in line with the invisible CO_2 beam. It shows where the head is, and Trace uses it to outline your job. [2]
`table`	The vector grid holds your sheet above the tray so the beam passes through without reflecting back. Cut-outs fall into the tray, where they become fuel. [2,5]
`extraction`	The fan pulls air from front to back, through the rear plenum and out of one 102 mm port. Epilog asks for 595–680 m³/h of extraction. At Saga it runs all the time. [2]
`control panel`	On the right shoulder: a 178 mm touch display, a joystick and Go/Stop. The ring around Go/Stop shows status; red means a drive has failed. [2,10]

Specifications

outside	1055 × 834 × 967 mm, 102 kg, on a built-in pedestal with wheels.[1]
work area	610 × 610 mm; material up to 254 mm thick.[1]
table	27.2 kg while it moves (autofocus moves it), 40.8 kg resting.[1,13]
laser	80 W CO_2, 10.6 µm, air-cooled metal/ceramic tube.[4]
lens	50.8 mm focal length, the standard lens.[1]
extraction	595–680 m³/h, one 102 mm port at the back.[2]
air	8 mm receptacle at the back, 2.1 bar maximum.[2]
power	Entry at the right side, rear corner.[2]

Go/Stop status ring



White
booting

Purple
loading

Multi
homing

Blue ↔ green
idle, ready

Green
running, tracing

Red
a drive failed

Red means a drive has failed: leave the machine and call staff.[2,10]

Run a job

Twelve steps from your file to a finished part, with what you should see at each one.

AT THE COMPUTER	1	**Design and check the file** Cut lines as vectors, engraving as filled areas or images. Know the material and its thickness before you start.	`vector` `raster`
	2	**Measure the sheet** Calipers, in two or three places. Use the thickest reading. [2]	`calipers`
	3	**Print to the laser** Epilog software: Auto Focus = Thickness. Split into processes (engrave, then cut) and enter the thickness in **every** process. [2]	`Auto Focus: Thickness`
	4	**Set speed and power** Start from the settings in Materials, or from the lab's saved settings for that material.	`speed` `power` `freq.`
AT THE MACHINE	5	**Turn it on** Turn the machine on and wait until the boot cycle is complete and the ring around Go/Stop shows idle (light blue to green). Check that air assist is on. Grid clear of offcuts, nothing on the rails. [2]	`Idle`
	6	**Place the sheet** Flat on the grid, inside 610 × 610 mm. Home is the upper-left (back-left) corner; job positions count from there. [1,2]	`home: back left`
	7	**Select the job** Job Menu: tap your file. It turns blue. [2]	`Job Menu`
	8	**Trace** Red dot on, then Trace. The head outlines the whole job. Make sure it stays on the sheet and clear of clips. [2]	`Pointer` `Trace`
	9	**Close the lid** The interlock blocks firing while it is open. [2]	`lid`
RUNNING AND AFTER	10	**Start and watch** The table rises to focus, then the job runs. Watch every second, closest during cuts. [2]	`Go/Stop`
	11	**Clear smoke, open lid** When the head parks, give the extraction a moment to clear the smoke. Then lift the lid. [2]	`Parking → Idle`
	12	**Clear the bed** Your part and your scraps. Look under the grid for anything still glowing. [5]	`tray`

DETAIL

Notes on the steps

Pause and resume

Go/Stop pauses: the beam stops at once; the head stops at the end of the raster line or the next vector node, so give it a moment. Lid up only when it has stopped. Lid down and Go/Stop resumes where it left off, and lines up only if nothing moved. [2]

Reset sends the head home and ends the run; the job stays in the list. [10]

Epilog software

The Epilog software prints the job to the laser. In it you choose Auto Focus (Off, Thickness or Plunger), split the job into processes, and set speed, power and frequency for each. At Saga, Auto Focus is always Thickness and the thickness goes into every process. [2]

Your designs

Start from Epilog’s settings for the material (Materials), and keep a note of what worked.

Focus, the red dot and home

Thickness focus, the red-dot pointer and the home position. Wrong focus is the most common cause of a bad result.

Saga method

Focus by thickness

In the software, Auto Focus = Thickness; enter the measured thickness in every process. When the job starts, the table rises so your surface sits at the focal point. Leave it blank and the machine focuses on the top of the grid: your sheet is then too close by its own thickness. [2]

placement

The red dot and Trace

The red dot is a visible laser in line with the invisible beam. With it on, Trace outlines the selected job over and over until you press Trace again. Use it to check placement before every job. [2]

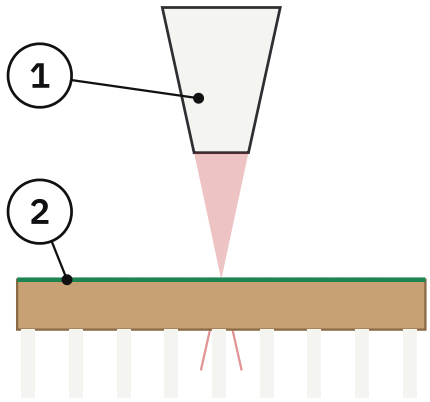
home position

Home and Reset

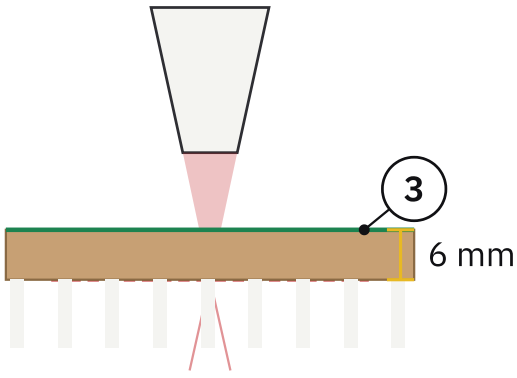
The head parks at its home position (back left) after a job. Reset sends it home from wherever it is. The status line shows Parking while it travels. [2,10]

FIGURE 1 · SECTION

Thickness entered vs. left blank



Thickness entered
surface on focus



Thickness blank
focus on the grid top

① **Lens and cone.** The beam narrows to its focal point a fixed distance below.

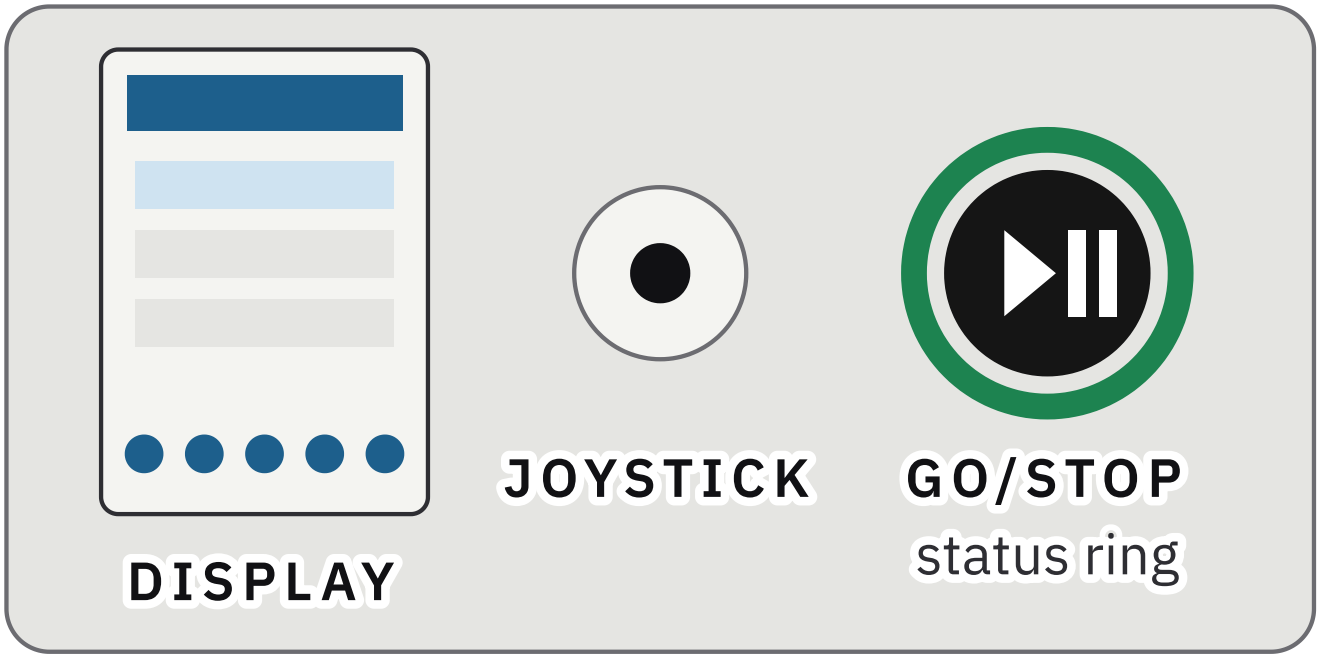
② **Thickness entered:** the table rises until your surface (green) sits on the focal plane (red).

③ **Blank:** it focuses on the grid top instead, and a 6 mm sheet stands 6 mm above focus. The spot is wider and weaker; the edges blur.

No thickness → it focuses on the top of the grid. [2] It is fault 2 in the 3D model.

FIGURE 2 · FROM ABOVE

Control panel



① **Display**, 178 mm: Job Menu, Focus, Pointer, Jog, Trace, status in the bottom-left corner.

② **Joystick**: jogs the head, or the table in the Focus menu.

③ **Go/Stop**: start, pause, resume. Its ring shows status.

Control panel: 178 mm display, joystick, Go/Stop. [2,10]

Plunger and gauge

Plunger and gauge. The machine also has a focus plunger and a manual focus gauge. Saga teaches thickness focus only. Never plunger-focus on the bare grid, and a gauge left down will crash into the work. [2]

Thick acrylic

Thick acrylic. For 6 mm and thicker, Epilog suggests moving focus about 2 mm closer to the lens for a cleaner edge. Only staff make this change. [3]

Calibration

Calibration. Focus and autofocus calibration have their own wizards on the panel. They belong to staff. [14]

Home, park and Reset

The head parks at its home position (back left) after a job. Reset sends it home from wherever it is. The status line shows Parking while it travels. [2,10]

Home is the upper-left (back-left) corner. [2]

Pre-flight checklist

A one-page checklist for every job, kept at the machine.

1 Before you load

- [ ] Signed off; staying at the machine for the whole job
- [ ] Material identified: name it, or read its safety data sheet. No PVC, no vinyl, no Kydex
- [ ] Thickness measured with calipers
- [ ] Heavy work is under 27.2 kg; otherwise ask staff
- [ ] Bed and tray clear of offcuts; nothing on the rails

2 At the panel

- [ ] Machine on, boot cycle complete: ring light blue to green
- [ ] Air assist on
- [ ] Job printed with Auto Focus = Thickness and the thickness in every process
- [ ] Job selected in the Job Menu (blue)
- [ ] Red dot on, Trace run: the outline stays on the sheet

3 Running

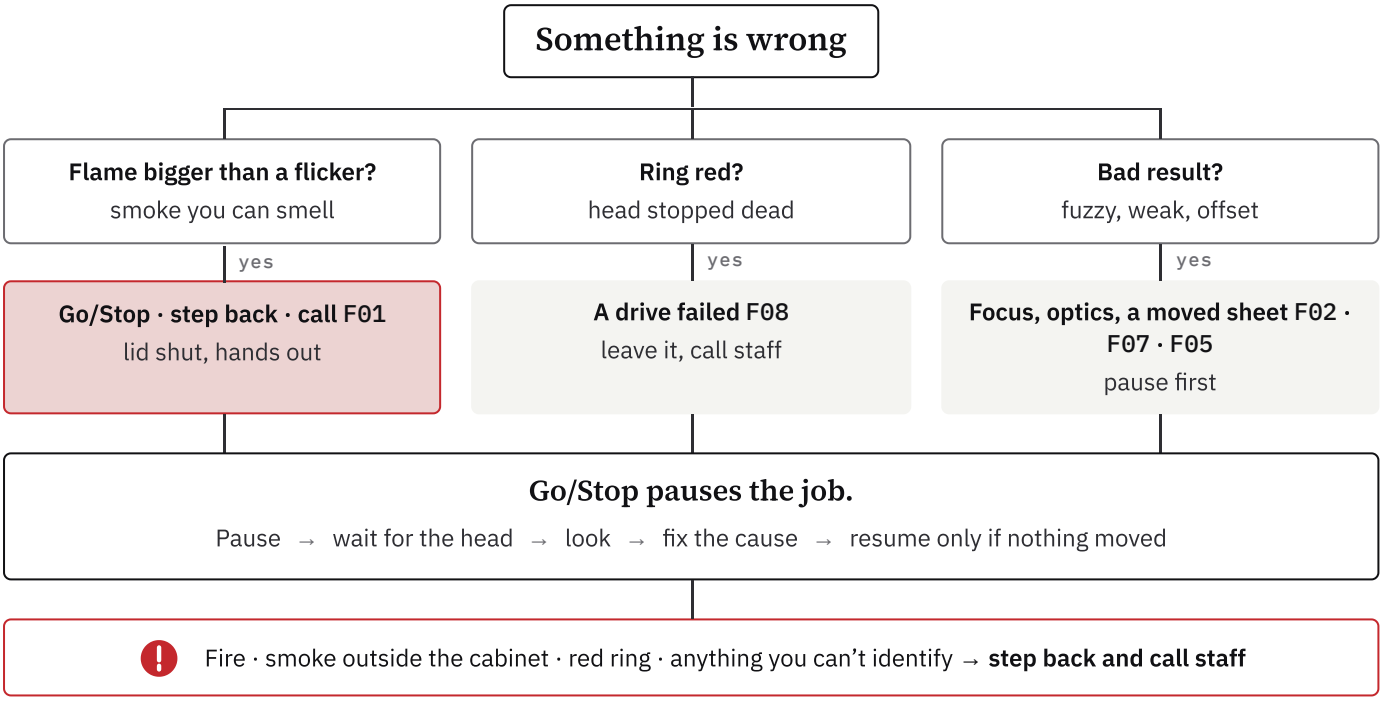
- [ ] Lid closed before Go/Stop
- [ ] You watch every second, closest during vector cuts
- [ ] Flame that grows: Go/Stop, step back, alert staff
- [ ] Go/Stop and wait for the head to stop before lifting the lid

4 After

- [ ] Smoke cleared before you open
- [ ] Part and scraps out; nothing glowing under the grid
- [ ] Anything odd written in the log and told to staff

Troubleshooting

Symptoms, causes and actions for ten faults. Pause the job before you diagnose.



INDEX

Symptoms

Each symptom gives the first action and links to the fault card and, where the 3D model stages it, its key.

Flames bigger than a flicker
Go/Stop. → F01

Fuzzy, wide mark or the cut doesn't go through
Go/Stop. → F02
Check thickness and settings first. → F07

Head moved with the lid up; rows missing
Close the lid. → F03

Smoke stays in the cabinet, or you can smell it
Go/Stop, keep the lid shut while it clears and tell staff. → F04

Two passes don't line up
Don't move the work during a pause. → F05

Glow or scraps under the grid
Take your scraps out. → F06

Lighter than last time, same settings
Check thickness and settings first. → F07

Ring red, head stopped
Leave it. → F08

Plastic with no label
Don't put it in. → F09

Your job isn't in the Job Menu
Check the printer chosen in the software and print again. → F10

When to step back and call staff

- Flame that grows, or any fire
- Smoke or smell outside the cabinet
- A red ring
- Anything you can't identify: material, noise, smell
- Optics, panels, the tray, calibration: always staff

REFERENCE

Ten faults

Each row gives the symptom, cause and action. Faults with a key are staged in the 3D model on The machine.

F01 Flame during a vector cut [2,5]

key 1

WHAT YOU SEE
Flame grows at the cut instead of a brief flicker.

WHY
Vector cutting puts the most heat into one place. Acrylic and wood flare up most; air assist reduces it but doesn't prevent it.

WHAT TO DO
Go/Stop. Don't lift the lid to look and don't reach in. Step back and get staff. Don't resume.

F02 No thickness entered [2]

key 2

WHAT YOU SEE
A wide, fuzzy mark; cuts don't go through.

WHY
With the thickness blank, autofocus focuses on the top of the grid, so the surface sits above focus by the sheet's thickness.

WHAT TO DO
Go/Stop. Measure with calipers, enter the thickness in every process, print again. More power doesn't fix focus.

F03 Lid opened mid-job [2]

key 3

WHAT YOU SEE
The beam stopped but the head kept moving; rows are missing.

WHY
The interlock stops the beam, not the motion.

WHAT TO DO
Close the lid. Rows done with it open are lost. Next time: Go/Stop, wait for the

head, then lift.

F04 Smoke lingers [2]

key 4

WHAT YOU SEE

Smoke hangs round the head, or you smell it at the lid or in the room.

WHY

Blocked plenum or filter, a leaking hose, or an extraction fault. The fan can still sound normal.

WHAT TO DO

Go/Stop, keep the lid shut while it clears and tell staff. Don’t run again until staff have checked the whole path.

F05 Second pass misses the first [2]

key 5

WHAT YOU SEE

Engraving and cut, or two passes, don’t line up.

WHY

The sheet moved while the lid was up. The machine works in its own coordinates and can’t know.

WHAT TO DO

Don’t move the work during a pause. If it moved, stop the job and set it up again with staff.

F06 Offcuts in the tray [2,5]

key 6

WHAT YOU SEE

Small pieces under the grid; a glow or smell after a cut.

WHY

Cut-outs that fall through are dry fuel right under the beam.

WHAT TO DO

Take your scraps out. If the tray is full or anything glows, tell staff; they empty it.

F07 Weaker mark than before [2]

key 7

WHAT YOU SEE

Same settings, lighter engraving or cuts that no longer go through.

WHY

Dirty or damaged optics lose power. Focus or settings can also be wrong.

WHAT TO DO

Check thickness and settings first. Then tell staff: optics are staff work, approved cleaner only.

F08 Red ring on Go/Stop [2,10]

key 8

WHAT YOU SEE

The head stops dead and the ring turns red.

WHY

A drive has failed.

WHAT TO DO

Leave it. Note what happened and call staff; they contact Epilog support.

F09 Unknown plastic[2,6,8]

key 9 —

WHAT YOU SEE

A sheet with no label: acrylic, polycarbonate, PVC?

WHY

PVC releases hydrogen chloride, which corrodes the machine and harms you. Polycarbonate cuts badly and discolours.

WHAT TO DO

Don’t put it in. Identify it from the label or the safety data sheet, or ask staff.

F10 Job not in the list[2]

—

WHAT YOU SEE

Your file doesn’t appear in the Job Menu.

WHY

The print didn’t reach the machine, or went to a different laser.

WHAT TO DO

Check the printer chosen in the software and print again. Network trouble is staff work.

Materials and settings

Identify the material first, then start from Epilog's 80 W settings and adjust from the result.

FIRST

Material identification

Epilog prohibits any material containing PVC or vinyl. Heated, it gives off hydrogen chloride, which corrodes the machine (voiding the warranty) and harms you. Kydex contains PVC. Neoprene's safety data sheet lists hydrogen chloride when heated. If you can't name a material, its safety data sheet tells you what is in it. You may bring any suitable material; ask staff about anything unknown. [2,6,7]

I Normally fine

- Wood, plywood, MDF, veneer
- Acrylic (cast or extruded)
- Paper, card, mat board
- Leather
- Cork, rubber
- Fabric, felt
- Glass, stone, tile (engrave only)
- Anodised aluminium (engrave only)

II Never

- PVC and vinyl, in any form
- Kydex and other PVC blends
- Neoprene / chloroprene (hydrogen chloride when heated)
- Anything you can't identify
- Bare metal for cutting

III Ask staff first

- Polycarbonate (cuts poorly, discolours)
- Coated or painted sheet
- Composites, foams, unknown plastics
- Anything over 27.2 kg

POLYCARBONATE

Allowed by Epilog, poor on a CO_2 laser

Epilog doesn’t forbid polycarbonate, but thin sheet cuts badly and discolours, and the fumes need good extraction. It looks like acrylic: if a clear sheet has no label, treat it as unknown. [2,8]

STARTING POINTS

Speed and power

Epilog lists its settings by material and wattage. These are the 80 W figures, for Saga’s laser.

MATERIAL	PROCESS	DPI / FREQ.	80 W STARTING POINT
Acrylic	Engrave text/clipart	300 DPI	100 s · 50 p
Acrylic	Engrave text/clipart	500 DPI	100 s · 40 p
Acrylic	Cut 3 mm	100 f	30 s · 100 p
Acrylic	Cut 6 mm	100 f	10 s · 100 p
Wood	Engrave text/clipart	500 DPI	80 s · 100 p
Wood	Thin veneer, cut	10 f	50 s · 40 p
Wood	Cut 3 mm	10 f	30 s · 100 p
Wood	Cut 6 mm	10 f	12 s · 100 p †
Cork	Engrave	500 DPI	100 s · 20 p
Anodised aluminium	Engrave text	500 DPI	100 s · 30 p

From the 80 W column of Epilog’s Fusion Edge suggested settings (CO_2). [3] s = speed %, p = power %, f = frequency. † Test this one on scrap before a real job. These are starting points: every sheet differs, so check the result and adjust. Cast acrylic engraves frosted; extruded cuts with a flame-polished edge. [3]

Epilog's CO_2 compatibility list

Epilog's list of materials a CO_2 laser engraves or cuts: acrylic, wood, paper, leather, cork, fabric, rubber, Delrin, Corian, mat board; engrave-only glass, ceramic, stone, tile and anodised aluminium. Bare metals only with a marking compound. [9]

Maintenance and reference

The published schedule, who does what, the numbers, the sources.

CALENDAR

Maintenance schedule

Epilog Support's preventive schedule for the Fusion Edge, split by who does it. Students do the first column; everything else is staff. [11]

I Every job (you)

- Take out your parts and scraps
- Look under the grid for anything glowing
- Report smells, weak marks, noises
- Leave the bed clear

II Weekly (staff)

- Clean the optics, approved cleaner only [2,11]
- Clean under the cutting table [11]
- Clean the Y guide rails [11]
- Wipe the interior [11]

III 6 months (staff)

- Remove and clean the exhaust plenum [11]
- Exhaust unit service [11]
- Fans and tube heat sinks clear [11]

IV 12 months

- Technician: full preventive service [11]

Cleaning and fire risk

Residue and debris build up on every job and become a fire hazard in their own right. Dirty optics lose power and are damaged by it. [2] Staff clean under the table weekly. [11]

Optics: staff only

Four mirrors, one window optic and one lens. Cleaning is staff work: approved optics cleaner (or reagent-grade alcohol) on cotton swabs, never isopropyl or hardware acetone. [2,11]

REFERENCE

Specifications and sources

Specifications

outside	1055 × 834 × 967 mm, 102 kg, on a built-in pedestal with wheels. [1]
work area	610 × 610 mm; material up to 254 mm thick. [1]
table	27.2 kg while it moves (autofocus moves it), 40.8 kg resting. [1,13]
laser	80 W CO_2, 10.6 µm, air-cooled metal/ceramic tube. [4]
lens	50.8 mm focal length, the standard lens. [1]
extraction	595–680 m³/h, one 102 mm port at the back. [2]
air	8 mm receptacle at the back, 2.1 bar maximum. [2]
power	Entry at the right side, rear corner. [2]

Sources

1. Epilog Laser — Fusion Edge product page and specifications (accessed 19 Oct 2026) — https://www.epiloglaser.com/laser-machines/fusion-edge-laser-cutter-engraver/
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5. Epilog Support KA-01237 — Reducing Fire Hazards with your Laser — https://support.epiloglaser.com/laser-machine/fusion-pro/usage-and-operation/faq/reducing-fire-hazards-with-your-laser/
6. Epilog Support KA-01212 — Material Safety and your Laser — https://support.epiloglaser.com/laser-machine/fusion-pro/usage-and-operation/faq/material-safety-and-your-laser/
7. Epilog FAQ — What materials are unsafe to engrave and/or cut with the laser? — https://www.epiloglaser.com/how-it-works/faq/laser-machine-unsafe-materials/
8. Epilog FAQ — Can you cut or engrave polycarbonate with a laser machine? — https://www.epiloglaser.com/how-it-works/faq/can-you-cut-engrave-polycarbonate-with-laser-machine/
9. Epilog Support KA-01508 — Material Compatibility for CO2 and Fiber Laser Engravers — https://support.epiloglaser.com/laser-machine/fusion-pro/usage-and-operation/faq/material-compatibility-for-co2-and-fiber-laser-engravers/
10. Epilog Support — Control Panel Tour, Fusion Pro, Edge, Maker & Galvo — https://support.epiloglaser.com/laser-machine/fusion-pro/getting-started/control-panel-tour/
11. Epilog Support KA-01555 — Cleaning Procedures, Fusion Maker, Edge and Pro — https://support.epiloglaser.com/laser-machine/fusion-pro/service-and-repair/cleaning/cleaning-procedures-fusion-maker-edge-and-pro/
12. Epilog Support KA-01672 — Optics Cleaning, Fusion Maker, Edge, Pro — https://support.epiloglaser.com/laser-machine/fusion-pro/service-and-repair/cleaning/optics-cleaning-fusion-maker-edge-pro/
13. Epilog Support KA-01583 — Engraver Table Weight Limits — https://support.epiloglaser.com/laser-machine/fusion-pro/usage-and-operation/system-requirements-setup/engraver-table-weight-limits/
14. Epilog Support KA-01684 — Autofocus Calibration Wizard (Fusion Edge) — https://support.epiloglaser.com/laser-machine/fusion-edge/calibrations/autofocus-calibration-wizard/
15. Epilog Support KA-01544 — Focus Calibrations, Fusion Maker, Edge and Pro — https://support.epiloglaser.com/laser-machine/fusion-edge/maintenance-cleaning-calibrations/focus-calibrations-fusion-maker-edge-and-pro/
16. Epilog Support KA-01619 — Lens Removal and Installation (Fusion Edge) — https://support.epiloglaser.com/laser-machine/fusion-edge/system-requirements-setup/lens-removal-and-installation/
17. Epilog Laser — official Fusion Edge 12/24/36 family product photograph (Edge 24: left foreground) — https://www.epiloglaser.com/assets/img/yootheme/cache/6c/fusion-edge-12-24-36-group-6c31c2a7.webp

Manuals are linked, never copied. Page numbers are the printed ones. The April 2021 Fusion Edge 12/24 manual matches this machine.

Self-check

Twelve questions. Passing is part of the sign-off.

Answers are printed below. Try each question before reading its answer.

1 Your 6 mm plywood cut leaves a wide, fuzzy line and doesn't go through. The settings are the lab's usual ones. What is the most likely cause, and what do you do?

> ANSWER · No thickness entered: it focused on the top of the grid, 6 mm below your surface. Go/Stop, measure, enter the thickness in every process, print again. See F02.

2 During a vector cut the flame grows beyond a flicker. What do you do, in order?

> ANSWER · Go/Stop. Don't lift the lid, don't reach in. Step back and get staff. Don't resume. How to stop, F01.

3 You want to check an engraving mid-job. Why must you press Go/Stop before lifting the lid?

> ANSWER · The lid stops the beam, not the head. Rows done with the lid up are lost, and the moving head is within reach. F03.

4 You paused, lifted the lid and nudged the sheet. What happens if you resume?

> ANSWER · It carries on in machine coordinates, so the rest lands offset. It only lines up if nothing moved. F05.

5 A friend gives you a grey plastic sheet with no label. What can you do with it?

> ANSWER · Nothing until it's identified (label or safety data sheet). If it is PVC, vinyl or Kydex, it never goes in. Materials, F09.

6 Why is PVC banned? Give two reasons.

> ANSWER · Heated, it releases hydrogen chloride: a corrosive gas that destroys the machine (and voids the warranty) and is harmful to breathe. Materials.

7 Where does the red dot land compared with the CO_2 beam, and what does Trace do?

> ANSWER · In line with it. Trace outlines the selected job with the red dot, repeatedly, so you can check placement. Focus.

8 The ring around Go/Stop turns red and the head stops. What do you do?

> ANSWER · A drive has failed. Leave it, note what happened, call staff. F08.

9 Smoke hangs in the cabinet after a cut and you can smell it at the lid. What do you do?

> ANSWER · Go/Stop, keep the lid shut while it clears, tell staff. Don't run again until they have checked the extraction path. F04.

10 What is the most you may put on the table for a job that uses autofocus, and why that figure?

> ANSWER · 27.2 kg: autofocus moves the table, and the lifting limit applies. 40.8 kg is only for resting. Rule 10.

11 After the job, what do you check before you leave?

> ANSWER · Part and scraps out, nothing glowing under the grid, bed clear, anything odd reported. Pre-flight.

12 A cut looks weaker than last week with the same settings and the right thickness. Who cleans the lens, and with what?

> ANSWER · Staff, with approved optics cleaner. Never isopropyl or hardware acetone. F07, Maintenance.